

**Happy birthday FTP !!**

File Transfer Protocol written by Abhay Bhusan officially turned 40 years old on 15th April 2011

**IPv4 addresses running out**

The Asia Pacific Network Information Centre (APNIC) will be rationing out its last block of IPv4 addresses as it ran out of them earlier than anticipated

# THE SCIENCE OF SOUND

## Apoorva Awasthy from Texas Instruments gives sound bytes

**Q** What are the trends that you observe in consumer audio systems these days?

- There's an increase in the demand for high quality audio
- There's also an increased demand for 2.1/5.1 sound bar systems, due to an increase in the adoption of flat panel displays
- Applications such as noise cancellation in mobile phones, higher loudness in portable audio systems are also getting popular
- Personalised home theatre systems with industry-standard decoding algorithms
- Wireless Speakers

**Q** How different are commercial audio systems (used in conference halls and theatres) from end users?

Commercial audio systems in theatres have higher number of channels, higher RMS output, higher sampling rates and higher fidelity. These systems have advanced room equalization algorithms resulting in better audience experience. The audio content in these systems is pre-encoded for multi-channel format such as Dolby and DTS.

Texas Instruments' (TI) end equipment solutions for audio applications integrate best-in-class systems expertise, providing increased application flexibility, higher performance and design longevity. TI is committed to providing the silicon, software, application knowledge and local technical support to help customers get to market faster.

**Q** What differentiates a good audio system from a conventional one?

There are a couple of features which differentiate a good audio system like:

- Higher fidelity
- Speaker response
- Higher sampling rate (>96 KHz). E.g –



Apoorva Awasthy, Business Development Manager, High Performance Analog (HPA), Texas Instruments India.

High-definition DVDs/Blu-ray discs

- Higher dynamic range - A good quality audio system should be able to accurately reproduce both quiet and loud sounds.
- Signal to noise ratio (SNR) – A high SNR is desired to reduce any background noise such as a hiss.
- Total Harmonic Distortion (THD) - A good system should have a very low THD.

**Q** Is it possible to enhance audio by means

of software? Any tips for our readers on how they do that?

Yes it is possible to enhance audio by means of software. The demand for portable consumer electronics is increasing rapidly. Obtaining good audio quality from a small portable consumer electronic device is a challenge that can be addressed using audio processing algorithms. These algorithms help in improving the audio quality of compressed audio files. A variety of algorithms are available today to process the